Inputs and Outputs of Input Validation

```
*IDLE Shell 3.12.1*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Hp\Desktop\SD - I (CW)\CW.py
Please enter the day of the survey in the format DD: 32
Out of range - values must be in the range 1 to 31.
Please enter the day of the survey in the format DD: 31
Please enter the month of the survey in the format MM: 13
Out of range - values must be in the range 1 to 12.
Please enter the day of the survey in the format DD: 31
Please enter the month of the survey in the format MM: 12
Please enter the year of the survey in the format YYYY: 2025
Out of range - values must range from 2000 to 2024.
Please enter the day of the survey in the format DD: 31
Please enter the month of the survey in the format MM: 12
Please enter the year of the survey in the format YYYY: 2024
Date entered: 31 / 12 / 2024

Would you like to load another dataset? (Y/N): Y

File is not available.
```

Inputs to get statical data of 15/06/2024

```
*IDLE Shell 3.12.1*
File  Edit  Shell  Debug  Options  Window  Help
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
Date entered: 15 / 6 / 2024

Would you like to load another dataset? (Y/N): Y
```

Outputs of 15/06/2024 statical data

```
IDLE Shell 3.12.1
File  Edit  Shell  Debug  Options  Window  Help


****************************************************
Data file selected is traffic_data15062024.csv
****************************************************

The total number of vehicles recorded for this date is 1037
The total number of trucks recorded for this date is 109
The total number of electric vehicles for this date is 368
The total number of two-wheeled vehicles for this date is 401
The total number of buses leaving Elm Avenue/Rabbit Road heading north is 15
The total number of vehicles passing through both junctions without turning left or right is 363
Percentage of all vehicles recorded that are trucks: 11%
Average number of bicycles per hour: 7
Total number of vehicles recorded as over the speed limit: 205
Total number of vehicles recorded through only Elm Avenue/Rabbit Road junction: 494
Total number of vehicles recorded through only Hanley Highway/Westway junction: 543
Percentage of vehicles through Elm Avenue/Rabbit Road that are scooters: 11%
Number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 39
Time or times of the peak (busiest) traffic hour (or hours) on Hanley Highway/Westway: Between 18:00 and 19:00
Total number of hours of rain on the selected date: 0

*******************************************************************************************

File written successfully!
```

Inputs to get statical data of 16/06/2024

```
IDLE Shell 3.12.1
File  Edit  Shell  Debug  Options  Window  Help

Please enter the day of the survey in the format DD: 16
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
Date entered: 16 / 6 / 2024

Would you like to load another dataset? (Y/N): Y
```

Outputs of 16/06/2024 statical data

```
IDLE Shell 3.12.1
File  Edit  Shell  Debug  Options  Window  Help

*****************************************************
Data file selected is traffic_data16062024.csv
*****************************************************

The total number of vehicles recorded for this date is 101
The total number of trucks recorded for this date is 11
The total number of electric vehicles for this date is 29
The total number of two-wheeled vehicles for this date is 29
The total number of buses leaving Elm Avenue/Rabbit Road heading north is 0
The total number of vehicles passing through both junctions without turning left or right is 38
Percentage of all vehicles recorded that are trucks: 11%
Average number of bicycles per hour: 0
Total number of vehicles recorded as over the speed limit: 20
Total number of vehicles recorded through only Elm Avenue/Rabbit Road junction: 52
Total number of vehicles recorded through only Hanley Highway/Westway junction: 49
Percentage of vehicles through Elm Avenue/Rabbit Road that are scooters: 6%
Number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 5
Time or times of the peak (busiest) traffic hour (or hours) on Hanley Highway/Westway: Between 1:00 and 2:00
Total number of hours of rain on the selected date: 9

*******************************************************************************************

File written successfully!
```

Inputs to get statical data of 21/06/2024

```
IDLE Shell 3.12.1
File  Edit  Shell  Debug  Options  Window  Help

Please enter the day of the survey in the format DD: 21
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
Date entered: 21 / 6 / 2024

Would you like to load another dataset? (Y/N): Y
```

Outputs of 21/06/2024 statical data

```
IDLE Shell 3.12.1
File  Edit  Shell  Debug  Options  Window  Help

*****************************************************
Data file selected is traffic_data21062024.csv
*****************************************************

The total number of vehicles recorded for this date is 1334
The total number of trucks recorded for this date is 138
The total number of electric vehicles for this date is 442
The total number of two-wheeled vehicles for this date is 503
The total number of buses leaving Elm Avenue/Rabbit Road heading north is 19
The total number of vehicles passing through both junctions without turning left or right is 494
Percentage of all vehicles recorded that are trucks: 10%
Average number of bicycles per hour: 10
Total number of vehicles recorded as over the speed limit: 250
Total number of vehicles recorded through only Elm Avenue/Rabbit Road junction: 651
Total number of vehicles recorded through only Hanley Highway/Westway junction: 683
Percentage of vehicles through Elm Avenue/Rabbit Road that are scooters: 10%
Number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 71
Time or times of the peak (busiest) traffic hour (or hours) on Hanley Highway/Westway: Between 18:00 and 19:00
Total number of hours of rain on the selected date: 389

*******************************************************************************************

File written successfully!
```

Inputs and Outputs when Existing program

```
IDLE Shell 3.12.1
File  Edit  Shell  Debug  Options  Window  Help
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
Date entered: 15 / 6 / 2024

Would you like to load another dataset? (Y/N): N

Exiting program...
>>>
```

Image of results text file: 15/06/2024

traffic_data15062024_results.txt

File Edit View

```
The total number of vehicles recorded for this date is 1037
The total number of trucks recorded for this date is 109
The total number of electric vehicles for this date is 368
The total number of two-wheeled vehicles for this date is 401
The total number of buses leaving Elm Avenue/Rabbit Road heading north is 15
The total number of vehicles passing through both junctions without turning left or right is 363
Percentage of all vehicles recorded that are trucks: 11%
Average number of bicycles per hour: 7
Total number of vehicles recorded as over the speed limit: 205
Total number of vehicles recorded through only Elm Avenue/Rabbit Road junction: 494
Total number of vehicles recorded through only Hanley Highway/Westway junction: 543
Percentage of vehicles through Elm Avenue/Rabbit Road that are scooters: 11%
Number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 39
Time or times of the peak (busiest) traffic hour (or hours) on Hanley Highway/Westway: Between 18:00 and 19:00
Total number of hours of rain on the selected date: 0
```

Ln 1, Col 1 | 1,066 characters | 100% | Windows (CRLF) | UTF-8

Image of results text file: 16/06/2024

traffic_data16062024_results.txt

File Edit View

```
The total number of vehicles recorded for this date is 101
The total number of trucks recorded for this date is 11
The total number of electric vehicles for this date is 29
The total number of two-wheeled vehicles for this date is 29
The total number of buses leaving Elm Avenue/Rabbit Road heading north is 0
The total number of vehicles passing through both junctions without turning left or right is 38
Percentage of all vehicles recorded that are trucks: 11%
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Percentage of vehicles through Elm Avenue/Rabbit Road that are scooters: 6%
Number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 5
Time or times of the peak (busiest) traffic hour (or hours) on Hanley Highway/Westway: Between 1:00 and 2:00
Total number of hours of rain on the selected date: 9
```

Ln 1, Col 1 | 1,053 characters | 100% | Windows (CRLF) | UTF-8

Image of results text file: 21/06/2024

traffic_data21062024_results.txt

File Edit View

```
The total number of vehicles recorded for this date is 1334
The total number of trucks recorded for this date is 138
The total number of electric vehicles for this date is 442
The total number of two-wheeled vehicles for this date is 503
The total number of buses leaving Elm Avenue/Rabbit Road heading north is 19
The total number of vehicles passing through both junctions without turning left or right is 494
Percentage of all vehicles recorded that are trucks: 10%
Average number of bicycles per hour: 10
Total number of vehicles recorded as over the speed limit: 250
Total number of vehicles recorded through only Elm Avenue/Rabbit Road junction: 651
Total number of vehicles recorded through only Hanley Highway/Westway junction: 683
Percentage of vehicles through Elm Avenue/Rabbit Road that are scooters: 10%
Number of vehicles recorded in the peak (busiest) hour on Hanley Highway/Westway: 71
Time or times of the peak (busiest) traffic hour (or hours) on Hanley Highway/Westway: Between 18:00 and 19:00
Total number of hours of rain on the selected date: 389
```

Ln 1, Col 1 | 1,069 characters | 100% | Windows (CRLF) | UTF-8